OpenNutri AI / Algorithm Layer

Midterm exposition | Retrieval, ranking, and feedback reuse

Current focus

- Multi-source paper discovery
- Bilingual EN/TR retrieval strategy
- Feedback-driven re-ranking for later runs

System claim at midterm

Retrieval, annotation, and feedback already work together as one operational loop.

Search, filtering, PDF acquisition, annotation, and later feedback reuse are all part of the current implementation.

Why This Layer Matters

The algorithmic layer exists to protect expert time and improve retrieval quality.

01

Distributed literature

Relevant food-composition papers are spread across multiple databases and local platforms.

02

High acquisition cost

Downloading every possible PDF is slow, noisy, and wastes expert review time.

03

Retrieval design

The system ranks metadata before acquisition and treats English and Turkish as separate pools.

Goal: spend acquisition and annotation effort only on papers with real food-composition potential.

Retrieval And Ranking Pipeline

The pipeline rejects weak candidates early and reserves PDF download for stronger papers.

Search Sources — **Search Gate** — **Metadata Filter** — **PDF Intake** — **Expert Annotation**

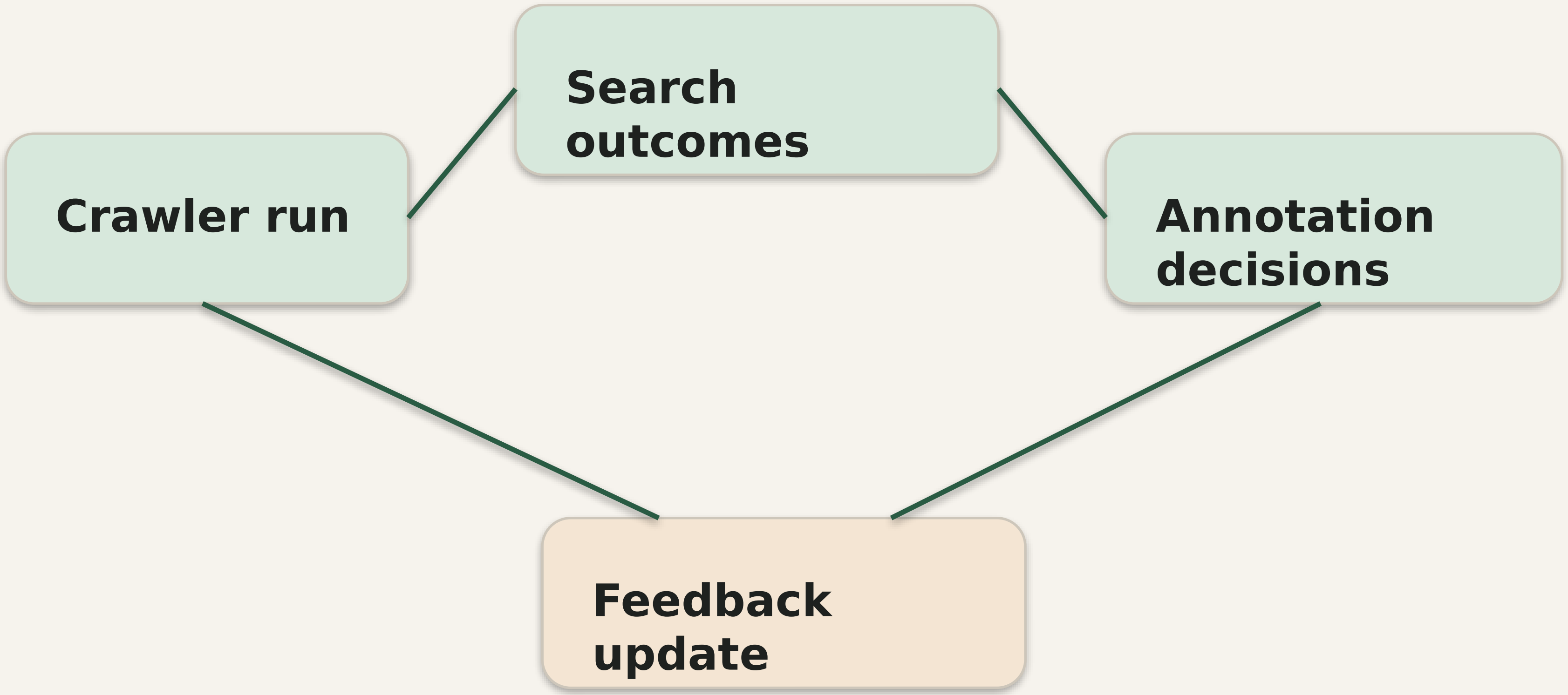
Signals inside metadata filter

- composition phrases
- food and nutrient terms
- unit patterns
- semantic similarity
- source and query priors
- feedback-weighted terms

Current implementation uses additive scoring: explicit lexical cues, embeddings, and learned feedback all contribute.

Implemented Feedback Loop

This loop is implemented now; it updates later crawler runs rather than only storing labels.



Update targets

- Latest visible label per user
- Positive, negative, and conflict sets
- Updated query and anchor phrases
- Updated concept and source priorities

Current state: a working batch feedback loop. It changes later runs, but it is not yet a trained classifier.

Current State And Scope Boundary

This slide helps you defend the project without overclaiming.

Implemented now

- Multi-source search across Europe PMC, OpenAlex, Semantic Scholar, and DergiPark
- Two-stage filtering before PDF download
- Embedding-supported metadata scoring
- Feedback-driven statistical term updates
- Paper-stock refill that closes the operational loop

Deferred to second semester

- Document segmentation
- LLM-assisted extraction
- Classifier training after label volume grows

Key midterm claim: retrieval, annotation, and feedback already work together as one system.